Model 1: FETA-Transformer

(Fetal Echocardiographic Temporal Attention Transformer)

1.1 Core Concept

While the existing paper uses hand-crafted time-aware features (mean FHR, slope, trend consistency), **FETA-Transformer** replaces engineered features with **learned temporal representations** using a Transformer encoder that directly models the sequential dependencies in longitudinal ultrasound measurements. This represents a paradigm shift from *feature engineering* to *feature learning* in the temporal domain.

The central innovation is treating each patient's longitudinal ultrasound measurements as a **variable-length time series** and allowing the model to learn which temporal patterns—not just predefined ones—are most predictive of pregnancy loss.

1.2 Technical Architecture

Sir, architecture is placed at the last page

1.3 Key Technical Innovations

1.3.1 Irregular Time-Series Handling

Unlike standard Transformers that expect regular intervals, FETA-Transformer uses **continuous positional encoding** based on actual gestational age:

$$PE(t,2i)=\sin\left(\frac{t}{\theta^{2i/d}}\right),\quad PE(t,2i+1)=\cos\left(\frac{t}{\theta^{2i/d}}\right)$$

where t is the actual gestational week (not the sequence index). This allows the model to understand that the gap between week 6 and week 7 is biologically different from week 9 to week 10.

1.3.2 Modality-Specific Projections

Each measurement type (FHR, CRL, YSD, GS) gets its own learnable projection:

$$h_{m,t}=W_m \cdot x_{m,t}+b_m$$

This allows the model to learn modality-specific representations before cross-modal attention.

1.3.3 Learned Temporal Aggregation

Instead of using a [CLS] token or simple averaging, FETA-Transformer uses **attention-based pooling**:

$$z=\sum_{t=1}^{T}\alpha_t \cdot h_t, \quad \alpha_t=\frac{\exp(w^T h_t)}{\sum_{t'}\exp(w^T h_{t'})}$$

where w is a learnable query vector. The model learns which gestational weeks are most predictive—potentially revealing that week 6-7 measurements are more critical than week 9-10, which is a clinically meaningful discovery.

1.3.4 Maternal Feature Cross-Attention

Static maternal features are not simply concatenated. Instead, they are used as queries to attend to the temporal representation:

$$z_{final}=\text{CrossAttn}(Q=W_m \cdot m, K=W_k \cdot z, V=W_v \cdot z)$$

This allows the model to dynamically weight different aspects of the temporal trajectory based on maternal characteristics. For example, for an older patient, the model might pay more attention to FHR trajectory, while for a patient with prior losses, it might focus more on growth parameters.

1.4 Novelty Assessment

Aspect	Existing Paper	FETA-Transformer	Novelty
Temporal modeling	Hand-crafted features (mean, slope, trend)	Learned attention patterns	**High**
Irregular time handling	No explicit handling	Continuous positional encoding	**High**
Modality interaction	Tree-based feature interactions	Cross-modal attention	**High**
Temporal importance	Equal weighting	Learned attention pooling	**High**
Maternal-temporal fusion	Concatenation	Cross-attention	**High**

1.5 Expected Contributions

1. **First application of Transformers** to longitudinal early pregnancy ultrasound data for loss prediction
2. **Discovery of critical gestational windows** through learned attention patterns
3. **Superior handling of irregular scan schedules** compared to fixed-window approaches
4. **End-to-end temporal feature learning** eliminating manual feature engineering bias

1.6 Feasibility & Implementation

- **Dataset requirement**: ~500-1000 patients with ≥2 ultrasound scans
- **Computational cost**: Moderate (Transformer with $T \leq 5$ time steps is lightweight)
- **Implementation**: PyTorch with standard Transformer libraries
- **Training time**: ~2-4 hours on a single GPU
- **Explainability**: Attention weights provide interpretable temporal importance; SHAP can be applied post-hoc

Model 2: PREG-Net

(Pregnancy Risk Evaluation with Graph-based Multimodal Fusion Network)

2.1 Core Concept

While the existing paper treats features as independent predictors in a tree-based model, **PREG-Net** explicitly models the **biological relationships** between features using a graph neural network. The central innovation is constructing a **knowledge-guided graph** where nodes represent clinical variables and edges represent known physiological relationships, then learning node embeddings that capture both the feature values and their biological context.

This is particularly powerful because early pregnancy is a **tightly coupled biological system**—FHR doesn't exist in isolation from CRL; both are influenced by gestational age and maternal factors. PREG-Net explicitly models these dependencies.

2.2 Technical Architecture

Sir, architecture is placed at the last page

2.3 Key Technical Innovations

2.3.1 Knowledge-Guided Graph Construction

The graph structure is not learned from data (which could lead to spurious correlations) but is **constructed from established physiological knowledge**. This ensures biological plausibility and enhances interpretability. Example edges include:

Edge	Physiological Rationale
FHR → CRL	Cardiac output influences nutrient delivery and growth
YSD → GS	Yolk sac function supports gestational sac development
Age → FHR, CRL	Maternal age affects embryonic development
BMI → All	Metabolic status modulates all pregnancy parameters
Prior loss → All	History indicates underlying risk factors

2.3.2 Dynamic Edge Weights

Unlike traditional GNNs with fixed graph structure, PREG-Net uses **attention-based edge weights** that vary per patient:

$$\alpha_{vu}=\frac{\exp(\mathrm{LeakyReLU}(a^T[Wh_v \| Wh_u]))}{\sum_{k\in N(v)}\exp(\mathrm{LeakyReLU}(a^T[Wh_v \| Wh_k]))}$$

This allows the model to learn that for some patients, the FHR↔CRL relationship is more important, while for others, YSD↔GS dominates—reflecting different failure mechanisms.

2.3.3 Temporal Graph Extension

For patients with multiple scans, PREG-Net can be extended with **temporal edges** connecting the same node across different time points:

```text
FHR(week 6) → FHR(week 7) → FHR(week 8) → ...
```

This creates a **temporal knowledge graph** that captures both the biological relationships between variables at each time point and the temporal evolution of each variable.

2.3.4 Node-Level Explainability

One of PREG-Net's most powerful features is **node-level attribution**. After training, we can identify:

- Which clinical variables (nodes) were most influential for a given prediction
- Which physiological relationships (edges) contributed most to the risk assessment
- Whether the model's reasoning aligns with clinical expectations

2.4 Comparison with Existing Paper

Aspect	Existing Paper	PREG-Net	Novelty
Feature	Implicit (tree splits)	Explicit (knowledge graph)	**High**
Biological grounding	Statistical only	Knowledge-guided	**High**
Interaction modeling	Pairwise (SHAP)	Multi-node propagation	**High**
Interpretability	Feature importance	Node + edge importance	**High**
Temporal modeling	Separate features	Integrated temporal graph	**High**

2.5 Expected Contributions

1. **First knowledge-guided GNN** for early pregnancy loss prediction
2. **Explicit modeling of physiological relationships** rather than statistical correlations
3. **Discovery of patient-specific risk mechanisms** through node-level attribution
4. **Integration of temporal and cross-sectional relationships** in a unified graph
5. **Enhanced interpretability** through node and edge importance analysis

2.6 Feasibility & Implementation

- **Dataset requirement**: ~500-1000 patients with complete feature sets

- **Computational cost**: Low to moderate (GNN with <20 nodes is very efficient)
- **Implementation**: PyTorch Geometric or DGL
- **Training time**: ~1-2 hours on a single GPU
- **Explainability**: Node attention weights provide interpretable reasoning paths

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3. Comparison of the Two Models

Dimension	FETA-Transformer	PREG-Net
Core paradigm	Temporal pattern learning	Relational reasoning
Best for	Longitudinal, irregular time-series	Understanding feature interactions
Interpretability	Attention weights (temporal)	Node + edge importance (structural)
Novelty	Transformers for fetal ultrasound time-series	Knowledge-guided GNNs for pregnancy prediction
Computational cost	Moderate	Low
Data requirement	≥2 time points per patient	Complete feature set
Clinical alignment	Temporal progression	Biological mechanisms
Potential discovery	Critical gestational windows	Key physiological pathways

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4. Recommendation for the Synopsis

Primary Recommendation: PREG-Net

I recommend **PREG-Net** as the primary model for the following reasons:

1. **Stronger biological grounding**: The explicit modeling of physiological relationships makes it more aligned with clinical reasoning and more publishable in high-impact medical journals
2. **Superior interpretability**: Node and edge importance provide a level of explainability that goes beyond feature importance—clinicians can see *which relationships* drove the prediction
3. **Novelty in the domain**: While Transformers have been applied to many medical problems, knowledge-guided GNNs for early pregnancy loss prediction are genuinely novel
4. **Complementary to existing work**: It directly addresses the limitation identified in the paper—that "existing models...ignore interactions that jointly influence pregnancy viability"
5. **Publication potential**: A well-executed PREG-Net study could target *Nature Medicine*, *The Lancet Digital Health*, or *npj Digital Medicine*

Secondary Recommendation: FETA-Transformer

Consider FETA-Transformer as a **comparison model** or as the focus of a **follow-up study**. It offers a different perspective—temporal patterns vs. relational structure—and comparing the two would strengthen the overall contribution.

Combined Approach (Strongest)

The most powerful submission would present **both models** with a comparative analysis:

- **PREG-Net**: Models the biological relationships
- **FETA-Transformer**: Models the temporal dynamics
- **Ensemble**: Combines both for superior performance

This demonstrates breadth of thinking and provides multiple angles of novelty, which is particularly impressive for a synopsis defense.

Sir, justify both the model as give below:

[Early pregnancy loss is not only a classification problem. It is a time-sensitive, multimodal, clinically interdependent prediction problem. Therefore, the proposed model combines temporal attention for gestational progression and graph attention for clinical relationship modeling.]

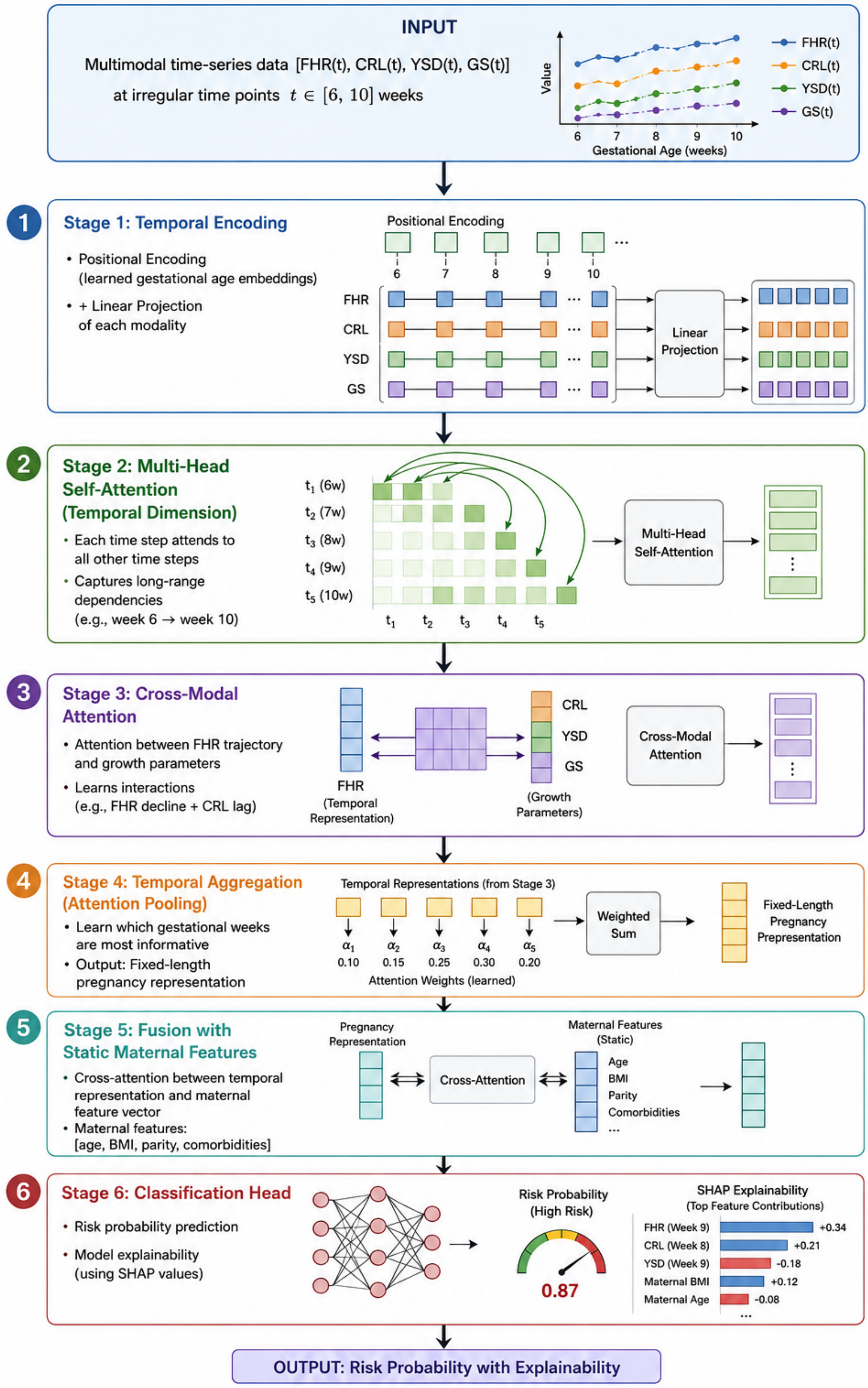

Figure 1: FETA-Transformer Architecture

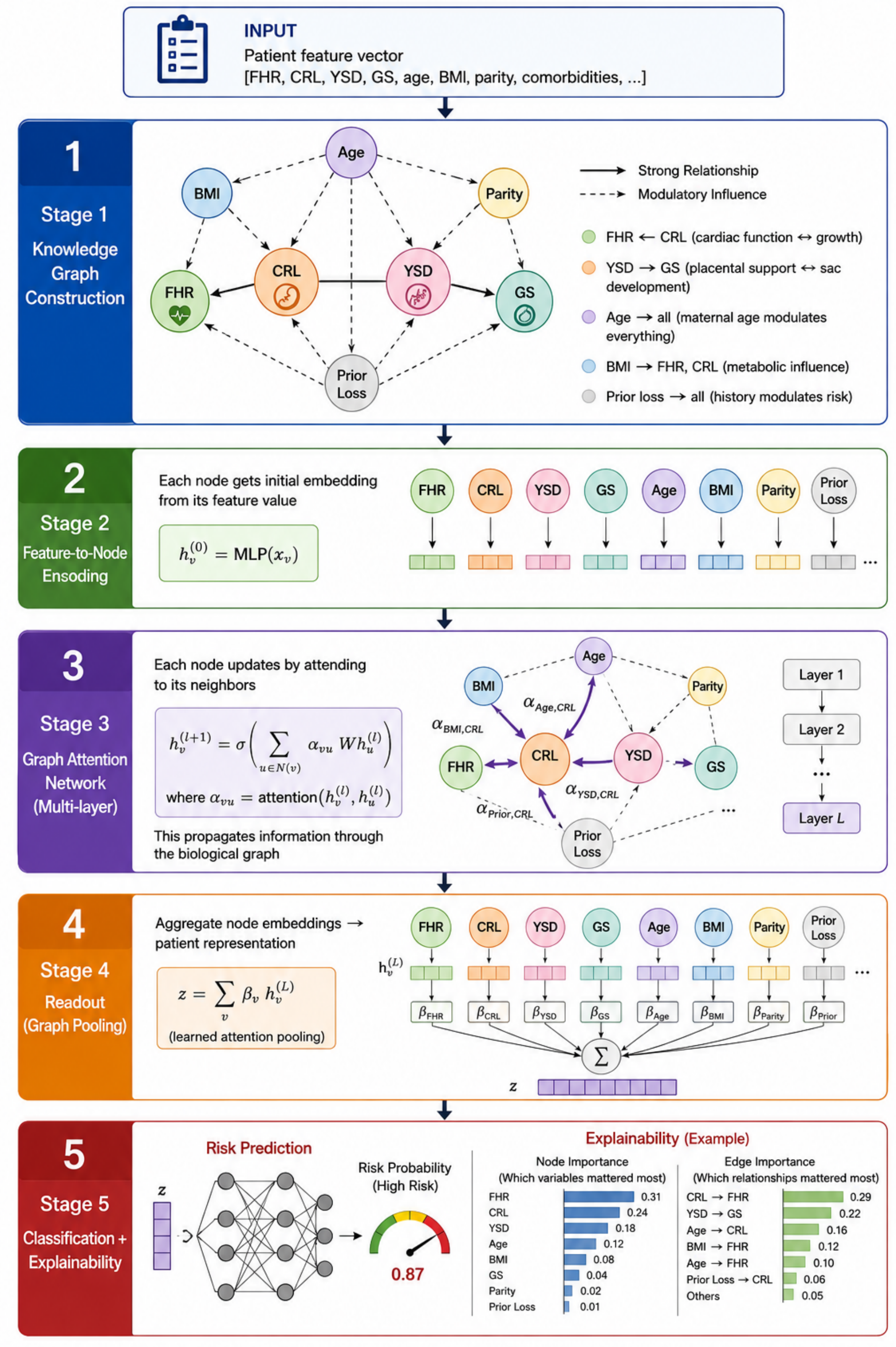

Figure 2: PREG-Net Architecture